MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
1000 GRADE SLAB
DESIGN OF HEAVY DUTY GRADE SLAB

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The purpose of this document is to analysis and design of **HEAVY DUTY GRADE SLAB**. In this document design & analysis of **HEAVY DUTY GRADE SLAB** have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

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3. SCOPE

The analysis has been carried out in Staad pro & designed by Excel Sheet. For grade slab strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. 15kN/m2 to 50 kN/m2 live load acting on grade slab in various staggered pattern.

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Grade slab: 30 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

LOAD 1 = DEAD LOAD (DL)

LOAD 2 = LOADING PATTERN-1 (LP-1)

LOAD 3 = LOADING PATTERN-2 (LP-2)

LOAD 4 = LOADING PATTERN-3 (LP-3)

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LOAD 5 = LOADING PATTERN-4 (LP-4)

LOAD 6 = LOADING PATTERN-5 (LP-5)

LOAD 7 = LOADING PATTERN-6 (LP-6)

4.3 Load combination

A. The following load combination are considered for member strength calculation.

101 1.4DL

102 1.4DL+1.7LP-1

103 1.4DL+1.7LP-2

104 1.4DL+1.7LP-3

105 1.4DL+1.7LP-4

106 1.4DL+1.7LP-5

107 1.4DL+1.7LP-6

B) The following load combination are considered for base pressure & sliding check.

201 1.0DL+1.0LP-1

202 1.0DL+1.0LP-2

203 1.0DL+1.0LP-3

204 1.0DL+1.0LP-4

205 1.0DL+1.0LP-5

206 1.0DL+1.0LP-6

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4.4 Minimum Reinforcing Steel (rebar) Size

The minimum rebar size for use as tension, compression reinforcement for grade slab shall be 8 mm

4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Grade slab	25

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 375 kN/m2 at 0.6m depth below EGL, refer Table 5 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

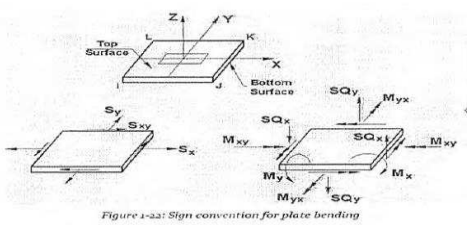
The Vertical Modulus of Subgrade Reaction has been considered as 33750 kN/m^2/m at 0.6m depth below EGL, refer Table 5 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

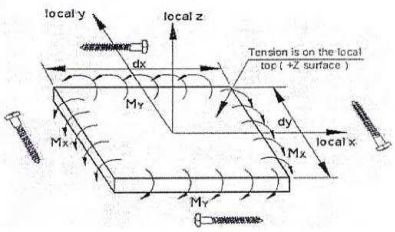
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5.1 Design of GRADE SLAB :





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; : MOMENT

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	15945	0.111	0.001	0	0	0	-1.876	-1.019	2.877	-4.753	-3.896	107 1.4DL+1.7LP-6
Min Qx	16368	-0.111	0.001	0	0	0	-1.876	-1.019	0	-1.876	-1.019	104 1.4DL+1.7LP-3
Max Qy	16434	-0.001	0.111	0	0	0	-1.019	-1.876	-2.877	-3.896	-4.753	104 1.4DL+1.7LP-3
Min Qy	16459	-0.001	-0.111	0	0	0	-1.019	-1.876	2.877	-3.896	-4.753	107 1.4DL+1.7LP-6
Max Sx	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Min Sx	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Max Sy	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Min Sy	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Max Sxy	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Min Sxy	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Max Mx	16860	0.002	-0.005	0	0	0	5.814	0.06	-0.109	5.923	0.169	102 1.4DL+1.7LP-1
Min Mx	16860	-0.002	0.005	0	0	0	-5.814	-0.06	0.109	-5.923	-0.169	103 1.4DL+1.7LP-2
Max My	15947	0.005	-0.002	0	0	0	0.06	5.814	-0.109	0.169	5.923	105 1.4DL+1.7LP-4
Min My	15947	-0.005	0.002	0	0	0	-0.06	-5.814	0.109	-0.169	-5.923	106 1.4DL+1.7LP-5
Max Mxy	16456	-0.039	-0.039	0	0	0	-0.454	-0.454	6.643	-7.097	-7.097	107 1.4DL+1.7LP-6
Min Mxy	16456	0.039	0.039	0	0	0	0.454	0.454	-6.643	7.097	7.097	104 1.4DL+1.7LP-3

PLATE CENTER STRESS ; SHEAR

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	15945	0.111	0.001	0	0	0	-1.876	-1.019	2.877	-4.753	-3.896	107 1.4DL+1.7LP-6
Min Qx	16368	-0.111	0.001	0	0	0	-1.876	-1.019	0	-1.876	-1.019	104 1.4DL+1.7LP-3
Max Qy	16434	-0.001	0.111	0	0	0	-1.019	-1.876	-2.877	-3.896	-4.753	104 1.4DL+1.7LP-3
Min Qy	16459	-0.001	-0.111	0	0	0	-1.019	-1.876	2.877	-3.896	-4.753	107 1.4DL+1.7LP-6
Max Sx	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Min Sx	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Max Sy	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Min Sy	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Max Sxy	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Min Sxy	16893	0	0	0	0	0	0	0	0	0	0	101 1.4DL
Max Mx	16860	0.002	-0.005	0	0	0	5.814	0.06	-0.109	5.923	0.169	102 1.4DL+1.7LP-1
Min Mx	16860	-0.002	0.005	0	0	0	-5.814	-0.06	0.109	-5.923	-0.169	103 1.4DL+1.7LP-2
Max My	15947	0.005	-0.002	0	0	0	0.06	5.814	-0.109	0.169	5.923	105 1.4DL+1.7LP-4
Min My	15947	-0.005	0.002	0	0	0	-0.06	-5.814	0.109	-0.169	-5.923	106 1.4DL+1.7LP-5
Max Mxy	16456	-0.039	-0.039	0	0	0	-0.454	-0.454	6.643	-7.097	-7.097	107 1.4DL+1.7LP-6
Min Mxy	16456	0.039	0.039	0	0	0	0.454	0.454	-6.643	7.097	7.097	104 1.4DL+1.7LP-3

Material Property :

Grade of concrete =		fc'30	
Grade of reinf. =		fy420	
Clear cover =	bottom face	25	mm
	top face	25	mm
Thicknes of slab =		150	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8357
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

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Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	5.92	top face	50	96	5.923	0.86	0.0021	199.3	225	8	223.49	200	252	4.15	4.966	0.055	tension control
Min Mx	-5.92	bottom face	50	96	5.923	0.86	0.0021	199.3		8	223.49	200	252	4.15	4.966	0.055	tension control
Max Mxy	-7.10	bottom face	50	96	7.097	1.03	0.0025	239.6		8	209.86	200	252	4.15	4.966	0.055	tension control
Min Mxy	7.10	top face	50	96	7.097	1.03	0.0025	239.6		8	209.86	200	252	4.15	4.966	0.055	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.111		16.65	4.753	0.003	0.096	85.568	88.4057	152.486	85.568	324	> 16.65	ok	64.18	φVc>Vu	not required
	Min Qx	0.111		16.65	1.876			87.773			87.773	326	> 16.65	ok	65.83	φVc>Vu	not required

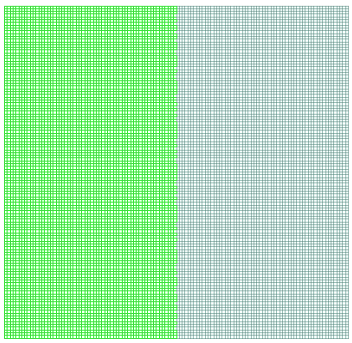
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	5.92	top face	50	96	5.923	0.86	0.0021	199.3	225	8	223.49	200	251.5	4.15	4.966	0.055	tension control
Min My	-5.92	bottom face	50	96	5.923	0.86	0.0021	199.3		8	223.49	200	251.5	4.15	4.966	0.055	tension control
Max Mxy	-7.10	bottom face	50	96	7.097	1.03	0.0025	239.6		8	209.86	200	251.5	4.15	4.966	0.055	tension control
Min Mxy	7.10	top face	50	96	7.097	1.03	0.0025	239.6		8	209.86	200	251.5	4.15	4.966	0.055	tension control

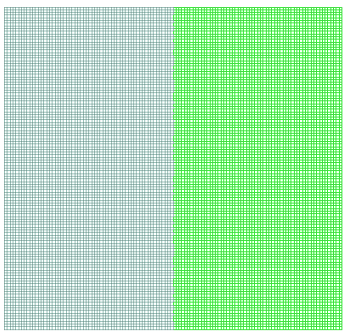
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.111		16.65	4.753	0.003	0.096	fc'85.57	fc'88.41	fc'152.5	85.568	90	> 16.65	ok	64.18	φVc>Vu	not required
	Min Qy	0.111		16.65	4.753			85.568			85.568	90	> 16.65	ok	64.18	φVc>Vu	not required

Loading patterns of grade slab

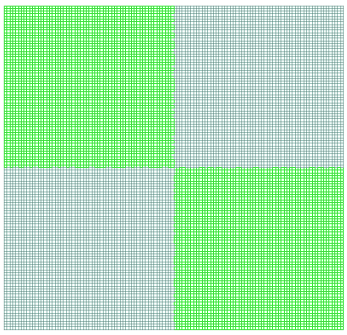
Load = 50kN/m2 Ks = 33750 kN/m2/m



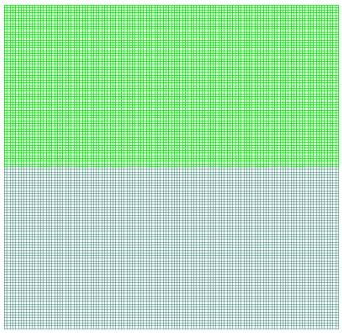
Loading pattern-1 of grade slab



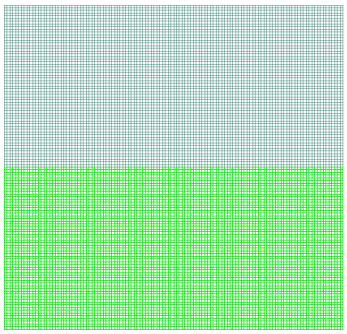
Loading pattern-2 of grade slab



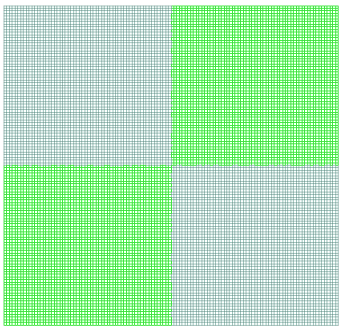
Loading pattern-3 of grade slab



Loading pattern-4 of grade slab



Loading pattern-5 of grade slab

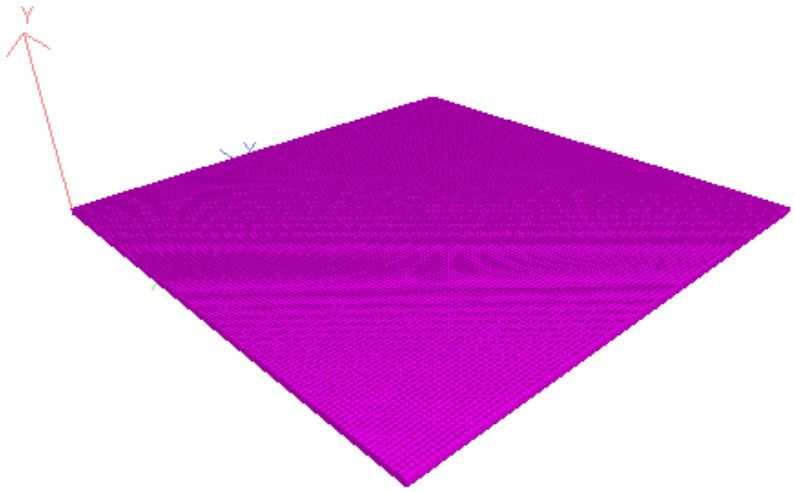


Loading pattern-6 of grade slab

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3D Rendered View of grade slab

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**8ϕT**	**200**	c/c
At Top	**8ϕT**	**200**	c/c
Along Z Direction :			
At Bottom	**8ϕT**	**200**	c/c
At Top	**8ϕT**	**200**	c/c

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5.2 Base pressure check of grade slab :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	1	201	0	53.681	0
Min Px	1	201	0	53.681	0
Max Py	15686	203	0	56.637	0
Min Py	15686	206	0	0.863	0
Max Pz	1	201	0	53.681	0
Min Pz	1	201	0	53.681	0

Gross Maximum Base pressure, p_{max} = 56.6 KN/m^2

Net Allowable Safe Bearing Capacity = 375 KN/m^2
Depth of soil above footing = 0.00 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 375 KN/m^2
> 56.6 KN/m^2
Hence OK

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1

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ATTACHMENT 1 STAAD INPUT & OUTPUT FILE FOR GRADE SLAB DESIGN

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```
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*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    FEB  8, 2021                *
*           Time=     9:27:56                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
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INPUT FILE: D:\SMPS JOBS\MAADEN GOLD\DESIGN\1000_GRADE SLAB\IFC\staad\Grade Slab DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 08-FEB-21
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 0 0 0; 2 12 0 0; 3 0 0 12; 4 12 0 12; 5 6 0 0; 6 6 0 12; 7 0 0 6; 8 12 0 6
    10. 9 6 0 6; 15 6.3 0 6; 21 6.6 0 6; 27 6.9 0 6; 33 7.2 0 6; 39 7.5 0 6
    11. 45 7.8 0 6; 51 8.1 0 6; 57 8.4 0 6; 63 8.7 0 6; 69 9 0 6; 75 9.3 0 6
    12. 81 9.6 0 6; 87 9.9 0 6; 93 10.2 0 6; 99 10.5 0 6; 105 10.8 0 6; 111 11.1 0 6
    13. 117 11.4 0 6; 123 11.7 0 6; 192 6 0 5.7; 375 6 0 5.4; 558 6 0 5.1; 741 6 0 4.8
    14. 924 6 0 4.5; 1107 6 0 4.2; 1290 6 0 3.9; 1473 6 0 3.6; 1656 6 0 3.3
    15. 1839 6 0 3; 2022 6 0 2.7; 2205 6 0 2.4; 2388 6 0 2.1; 2571 6 0 1.8
    16. 2754 6 0 1.5; 2937 6 0 1.2; 3120 6 0 0.900001; 3303 6 0 0.600001
    17. 3486 6 0 0.300002; 3732 0.3 0 6; 3738 0.6 0 6; 3744 0.9 0 6; 3750 1.2 0 6
    18. 3756 1.5 0 6; 3762 1.8 0 6; 3768 2.1 0 6; 3774 2.4 0 6; 3780 2.7 0 6
    19. 3786 3 0 6; 3792 3.3 0 6; 3798 3.6 0 6; 3804 3.9 0 6; 3810 4.2 0 6
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    22. 8482 6 0 10.2; 8665 6 0 9.9; 8848 6 0 9.6; 9031 6 0 9.3; 9214 6 0 9
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1223. ELEMENT
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1271. MEMBER
1272. END GROUP DEFINITION
1273. ELEMENT PROPERTY
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1277. E 2.17185E+007
1278. POISSON 0.17
1279. DENSITY 25
1280. ALPHA 1E-005
1281. DAMP 0.05
1282. TYPE CONCRETE
1283. STRENGTH FCU 27579
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1285. CONSTANTS
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1289. ************************************************************************
1290. SUPPORTS
1291. _RAFT PLATE MAT DIRECT Y SUBGRADE 33750
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1294. SELFWEIGHT Y -1
1295. ************************************************************************
1296. LOAD 2 LOADING PATTERN-1 (LP-1)
1297. ELEMENT LOAD
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1299. 15233 TO 15252 15254 TO 15273 15275 TO 15294 15296 TO 15315 15317 TO 15336 -
1300. 15338 TO 15357 15359 TO 15378 15380 TO 15399 15401 TO 15420 15422 TO 15441 -
1301. 15443 TO 15462 15464 TO 15483 15485 TO 15504 15506 TO 15525 15527 15529 -
1302. 15531 15533 15535 15537 15539 15541 15543 15545 15547 15549 15551 15553 -
1303. 15555 15557 15559 15561 15563 15564 PR -50
1304. 15572 15574 15576 15578 15580 15582 15584 15586 15588 15590 15592 15594 15596 -
1305. 15598 15600 15602 15604 15606 15608 15609 15611 TO 15630 15632 TO 15651 -
1306. 15653 TO 15672 15674 TO 15693 15695 TO 15714 15716 TO 15735 15737 TO 15756 -
1307. 15758 TO 15777 15779 TO 15798 15800 TO 15819 15821 TO 15840 15842 TO 15861 -
1308. 15863 TO 15882 15884 TO 15903 15905 TO 15924 15926 TO 15945 15947 TO 15966 -
1309. 15968 TO 16007 PR -50
1310. ***********************************
1311. LOAD 3 LOADING PATTERN-2 (LP-2)
1312. ELEMENT LOAD
1313. 16014 16016 16018 16020 16022 16024 16026 16028 16030 16032 16034 16036 16038 -
1314. 16040 16042 16044 16046 16048 16050 16052 TO 16071 16073 TO 16092 -
1315. 16094 TO 16113 16115 TO 16134 16136 TO 16155 16157 TO 16176 16178 TO 16197 -
1316. 16199 TO 16218 16220 TO 16239 16241 TO 16260 16262 TO 16281 16283 TO 16302 -
1317. 16304 TO 16323 16325 TO 16344 16346 TO 16365 16367 TO 16386 16388 TO 16407 -
1318. 16409 TO 16428 16430 TO 16450 PR -50
1319. 16456 TO 16474 16476 TO 16495 16497 TO 16516 16518 TO 16537 16539 TO 16558 -
1320. 16560 TO 16579 16581 TO 16600 16602 TO 16621 16623 TO 16642 16644 TO 16663 -
1321. 16665 TO 16684 16686 TO 16705 16707 TO 16726 16728 TO 16747 16749 TO 16768 -
1322. 16770 TO 16789 16791 TO 16810 16812 TO 16831 16833 TO 16852 16854 16856 -
1323. 16858 16860 16862 16864 16866 16868 16870 16872 16874 16876 16878 16880 -
1324. 16882 16884 16886 16888 16890 16892 16893 PR -50
1325. ***********************************
1326. LOAD 4 LOADING PATTERN-3 (LP-3)
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1327. ELEMENT LOAD
1328. 15128 TO 15147 15149 TO 15168 15170 TO 15189 15191 TO 15210 15212 TO 15231 -
1329. 15233 TO 15252 15254 TO 15273 15275 TO 15294 15296 TO 15315 15317 TO 15336 -
1330. 15338 TO 15357 15359 TO 15378 15380 TO 15399 15401 TO 15420 15422 TO 15441 -
1331. 15443 TO 15462 15464 TO 15483 15485 TO 15504 15506 TO 15525 15527 15529 -
1332. 15531 15533 15535 15537 15539 15541 15543 15545 15547 15549 15551 15553 -
1333. 15555 15557 15559 15561 15563 15564 PR -50
1334. 16014 16016 16018 16020 16022 16024 16026 16028 16030 16032 16034 16036 16038 -
1335. 16040 16042 16044 16046 16048 16050 16052 TO 16071 16073 TO 16092 -
1336. 16094 TO 16113 16115 TO 16134 16136 TO 16155 16157 TO 16176 16178 TO 16197 -
1337. 16199 TO 16218 16220 TO 16239 16241 TO 16260 16262 TO 16281 16283 TO 16302 -
1338. 16304 TO 16323 16325 TO 16344 16346 TO 16365 16367 TO 16386 16388 TO 16407 -
1339. 16409 TO 16428 16430 TO 16450 PR -50
1340. ************************************
1341. LOAD 5 LOADING PATTERN-4 (LP-4)
1342. ELEMENT LOAD
1343. 15128 TO 15147 15149 TO 15168 15170 TO 15189 15191 TO 15210 15212 TO 15231 -
1344. 15233 TO 15252 15254 TO 15273 15275 TO 15294 15296 TO 15315 15317 TO 15336 -
1345. 15338 TO 15357 15359 TO 15378 15380 TO 15399 15401 TO 15420 15422 TO 15441 -
1346. 15443 TO 15462 15464 TO 15483 15485 TO 15504 15506 TO 15525 15527 15529 -
1347. 15531 15533 15535 15537 15539 15541 15543 15545 15547 15549 15551 15553 -
1348. 15555 15557 15559 15561 15563 15564 PR -50
1349. 16456 TO 16474 16476 TO 16495 16497 TO 16516 16518 TO 16537 16539 TO 16558 -
1350. 16560 TO 16579 16581 TO 16600 16602 TO 16621 16623 TO 16642 16644 TO 16663 -
1351. 16665 TO 16684 16686 TO 16705 16707 TO 16726 16728 TO 16747 16749 TO 16768 -
1352. 16770 TO 16789 16791 TO 16810 16812 TO 16831 16833 TO 16852 16854 16856 -
1353. 16858 16860 16862 16864 16866 16868 16870 16872 16874 16876 16878 16880 -
1354. 16882 16884 16886 16888 16890 16892 16893 PR -50
1355. ************************************
1356. LOAD 6 LOADING PATTERN-5 (LP-5)
1357. ELEMENT LOAD
1358. 15572 15574 15576 15578 15580 15582 15584 15586 15588 15590 15592 15594 15596 -
1359. 15598 15600 15602 15604 15606 15608 15609 15611 TO 15630 15632 TO 15651 -
1360. 15653 TO 15672 15674 TO 15693 15695 TO 15714 15716 TO 15735 15737 TO 15756 -
1361. 15758 TO 15777 15779 TO 15798 15800 TO 15819 15821 TO 15840 15842 TO 15861 -
1362. 15863 TO 15882 15884 TO 15903 15905 TO 15924 15926 TO 15945 15947 TO 15966 -
1363. 15968 TO 16007 PR -50
1364. 16014 16016 16018 16020 16022 16024 16026 16028 16030 16032 16034 16036 16038 -
1365. 16040 16042 16044 16046 16048 16050 16052 TO 16071 16073 TO 16092 -
1366. 16094 TO 16113 16115 TO 16134 16136 TO 16155 16157 TO 16176 16178 TO 16197 -
1367. 16199 TO 16218 16220 TO 16239 16241 TO 16260 16262 TO 16281 16283 TO 16302 -
1368. 16304 TO 16323 16325 TO 16344 16346 TO 16365 16367 TO 16386 16388 TO 16407 -
1369. 16409 TO 16428 16430 TO 16450 PR -50
1370. ************************************
1371. LOAD 7 LOADING PATTERN-6 (LP-6)
1372. ELEMENT LOAD
1373. 15572 15574 15576 15578 15580 15582 15584 15586 15588 15590 15592 15594 15596 -
1374. 15598 15600 15602 15604 15606 15608 15609 15611 TO 15630 15632 TO 15651 -
1375. 15653 TO 15672 15674 TO 15693 15695 TO 15714 15716 TO 15735 15737 TO 15756 -
1376. 15758 TO 15777 15779 TO 15798 15800 TO 15819 15821 TO 15840 15842 TO 15861 -
1377. 15863 TO 15882 15884 TO 15903 15905 TO 15924 15926 TO 15945 15947 TO 15966 -
1378. 15968 TO 16007 PR -50
1379. 16456 TO 16474 16476 TO 16495 16497 TO 16516 16518 TO 16537 16539 TO 16558 -
1380. 16560 TO 16579 16581 TO 16600 16602 TO 16621 16623 TO 16642 16644 TO 16663 -
1381. 16665 TO 16684 16686 TO 16705 16707 TO 16726 16728 TO 16747 16749 TO 16768 -
1382. 16770 TO 16789 16791 TO 16810 16812 TO 16831 16833 TO 16852 16854 16856 -
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1383. 16858 16860 16862 16864 16866 16868 16870 16872 16874 16876 16878 16880 -
1384. 16882 16884 16886 16888 16890 16892 16893 PR -50
1385. ******************************************************************************
1386. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
1387. ******************************************************************************
1388. LOAD COMB 101 1.4DL
1389. 1 1.4
1390. LOAD COMB 102 1.4DL+1.7LP-1
1391. 1 1.4 2 1.7
1392. LOAD COMB 103 1.4DL+1.7LP-2
1393. 1 1.4 3 1.7
1394. LOAD COMB 104 1.4DL+1.7LP-3
1395. 1 1.4 4 1.7
1396. LOAD COMB 105 1.4DL+1.7LP-4
1397. 1 1.4 5 1.7
1398. LOAD COMB 106 1.4DL+1.7LP-5
1399. 1 1.4 6 1.7
1400. LOAD COMB 107 1.4DL+1.7LP-6
1401. 1 1.4 7 1.7
1402. ******************************************************************************
1403. ***UN-FACTORED LOAD COMBINATIONS FOR STABILITY CHECK
1404. ******************************************************************************
1405. LOAD COMB 201 1.0DL+1.0LP-1
1406. 1 1.0 2 1.0
1407. LOAD COMB 202 1.0DL+1.0LP-2
1408. 1 1.0 3 1.0
1409. LOAD COMB 203 1.0DL+1.0LP-3
1410. 1 1.0 4 1.0
1411. LOAD COMB 204 1.0DL+1.0LP-4
1412. 1 1.0 5 1.0
1413. LOAD COMB 205 1.0DL+1.0LP-5
1414. 1 1.0 6 1.0
1415. LOAD COMB 206 1.0DL+1.0LP-6
1416. 1 1.0 7 1.0
1417. ******************************************************************************
1418. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 1681 NUMBER OF MEMBERS 0
NUMBER OF PLATES 1600 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1681

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1679/ 66/ 201 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 5043
TOTAL LOAD COMBINATION CASES = 13 SO FAR.
SIZE OF STIFFNESS MATRIX = 1014 DOUBLE KILO-WORDS

REQRD/AVAIL. DISK SPACE = 31.7/ 92901.0 MB

1419. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= FEB 8,2021 TIME= 9:28: 6 ****

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